

Multimodal Foundation Model for Real-Time Drone Detection & Intent Inference: Expanded Explanation

Introduction

The idea of a multimodal foundation model for UAV (drone) detection and intent inference addresses the persistent challenges in drone surveillance: reliably detecting small, fast-moving objects in cluttered or adverse environments, and inferring whether their intent is benign or malicious. Recent advances in AI/ML—especially transformer-based architectures and multimodal sensor fusion—offer promising solutions to these problems by integrating diverse data streams (visual, radar, RF, acoustic) and leveraging powerful attention mechanisms for robust, real-time performance (Al-IQubaydhi et al., 2024; Larrat & Sales, 2025; Chen et al., 2024; Luan et al., 2025).

Why Multimodal Fusion?

Single-modality systems (e.g., vision-only or radar-only) are limited by environmental factors:

- Visual sensors struggle with occlusion, poor lighting, or weather.
- Radar/RF/acoustic sensors can be confounded by noise or similar signatures from birds/vehicles (Al-IQubaydhi et al., 2024; Seidaliyeva et al., 2023).
- Small drones are especially hard to detect due to their size and low signal strength (Tan et al., 2024; Cheng et al., 2024).

Multimodal fusion combines complementary strengths:

- Visual: shape, color, movement
- Radar: range, velocity
- RF: communication patterns
- Acoustic: propeller/frequency signatures

By fusing these modalities using deep learning—especially transformers with self-attention—models can learn richer representations and become more robust to missing/noisy data (Larrat & Sales, 2025; Zhao et al., 2024; Luan et al., 2025).

Core Technical Approach

1. Model Architecture

- **Multimodal Transformer Backbone:** Integrates features from radar, video (RGB/IR), audio, and potentially RF streams. Each modality is processed through its own encoder; features are fused via cross-attention/self-attention layers (Larrat & Sales, 2025; Chen et al., 2024; Zhao et al., 2024).
- Example: A model fusing radar/image/audio achieved >98% accuracy and real-time speeds (>40 FPS) on challenging aerial datasets (Larrat & Sales, 2025).
- Feature fusion enables the model to focus on salient cues across modalities—even when one sensor is degraded.

2. Diffusion Models for Data Augmentation & Robustness

- Diffusion models can generate synthetic training samples for rare events (e.g., malicious drone behaviors), improving generalization to unseen scenarios (Li et al., 2024).
- They also help mitigate class imbalance—a common issue in intent recognition datasets.

3. Reinforcement Learning (RL) Post-training

- RL fine-tuning allows the system to adapt its detection and classification strategies based on feedback from real-world deployments.
- For example, RL can optimize the trade-off between false positives (unnecessary alarms) and false negatives (missed threats), learning from operator feedback or simulated adversarial scenarios (Rahman et al., 2024; Bai et al., 2023).
- RL-based approaches have shown improved generalization to new drone types/environments compared to static supervised models.

4. Intent Inference

- Beyond detection (“is there a drone?”), the system classifies intent using temporal patterns across modalities:
 - Benign: delivery routes, survey patterns
 - Malicious: erratic flight paths near sensitive areas
- Cross-modal transformers excel at modeling such temporal dependencies and context (Li et al., 2024).

Expected Contributions

1. **Robust All-weather Detection:** By leveraging multiple sensors, the system maintains high accuracy even if one modality fails or is noisy (e.g., fog blocks cameras but not radar).
2. **Small Object & Clutter Handling:** Attention mechanisms in transformers improve detection of small drones amidst cluttered backgrounds—a major limitation of older CNN-based methods (Tan et al., 2024; Ye et al., 2023; Tan et al., 2025).
3. **Intent Classification:** The model not only detects drones but also infers likely intent in real time—crucial for prioritizing responses in security-critical zones (Li et al., 2024).

4. **Reduced False Alarms:** Sensor fusion and advanced ML reduce false positives compared to single-modality systems—a key operational requirement (Svanström et al., 2020; Svanström et al., 2022).
5. **Real-time Performance:** Efficient transformer designs achieve high frame rates suitable for live deployment on embedded hardware (Larrat & Sales, 2025; Ye et al., 2023).

Feasibility & State-of-the-Art Support

Recent research demonstrates that:

- Multimodal transformer models outperform single-modality baselines in both accuracy and robustness for UAV detection/classification tasks (Larrat & Sales, 2025; Chen et al., 2024; Zhao et al., 2024).
- Lightweight transformer variants enable deployment on edge devices with limited compute resources while maintaining real-time performance (Ye et al., 2023; Tan et al., 2025).
- RL post-training further enhances adaptability to new environments/threats without extensive retraining (Rahman et al., 2024; Bai et al., 2023).

Datasets combining visible/thermal video, audio, radar/RF signals are increasingly available for benchmarking such systems (Svanström et al., 2020). Public codebases exist for many state-of-the-art models cited here.

Key References

Core Aspect	Supporting Papers
Multimodal transformer fusion	(Larrat & Sales, 2025), (Chen et al., 2024), (Zhao et al., 2024), (Luan et al., 2025), (Azeem et al., 2024)
Small object/clutter handling	(Tan et al., 2024), (Ye et al., 2023), (Tan et al., 2025), (Liu et al., 2025)
Intent inference	(Li et al., 2024), (Li et al., 2024)
RL post-training/generalization	(Rahman et al., 2024), (Bai et al., 2023)
Real-time/edge deployment	(Larrat & Sales, 2025), (Ye et al., 2023), (Tan et al., 2025)
Sensor fusion benefits	(Al-IQubaydhi et al., 2024), (Seidaliyeva et al., 2023), (Svanström et al., 2020), (Svanström et al., 2022)

Summary Table: Capabilities vs. Challenges

Challenge Addressed	How This Approach Helps	Citations
Small object detection	Attention-based feature fusion across modalities	(Tan et al., 2024), (Ye et al., 2023), (Tan et al., 2025), (Liu et al., 2025)
Adverse weather/noise	Redundant sensing; robust transformer architectures	(Al-IQubaydhi et al., 2024), (Seidaliyeva et al., 2023), (Svanström et al., 2020)
False alarms	Cross-modal context reduces spurious detections	(Svanström et al., 2020), (Svanström et al., 2022)
Intent prediction	Temporal modeling + multimodal context	(Li et al., 2024), (Li et al., 2024)
Generalization to new threats	RL post-training; diffusion-based augmentation	(Rahman et al., 2024), (Bai et al., 2023), (Li et al., 2024)

FIGURE 1 Summary of how multimodal foundation models address UAV detection challenges.

Conclusion

A multimodal foundation model using transformer/diffusion architectures—with RL-based adaptation—offers a feasible path toward robust, real-time UAV detection and intent inference. Such systems promise significant improvements over current single-modality or unimodal deep learning approaches by reducing false alarms, improving resilience under challenging conditions, enabling intent classification, and supporting practical deployment on embedded platforms—all validated by recent research trends and empirical results in the field.

Key next steps:

- Expand public datasets with synchronized multi-sensor data.
- Develop open-source benchmarks for intent inference.
- Explore lightweight transformer/diffusion variants for edge deployment.

References:

- (Al-IQubaydhi et al., 2024): Deep learning for unmanned aerial vehicles detection: A review
- (Larrat & Sales, 2025): A Multimodal Transformer Approach for UAV Detection and Aerial Object Recognition Using Radar, Audio, and Video Data
- (Tan et al., 2024): A Real-Time Unmanned Aerial Vehicle (UAV) Aerial Image Object Detection Model
- (Chen et al., 2024): Drone-Based Visible–Thermal Object Detection with Transformers and Prompt Tuning
- (Ye et al., 2023): Real-Time Object Detection Network in UAV-Vision Based on CNN and Transformer
- (Zhao et al., 2024): Deep Multimodal Data Fusion
- (Luan et al., 2025): Fast Identification and Detection Algorithm for Maneuverable Unmanned Aircraft Based on Multimodal Data Fusion
- (Rahman et al., 2024): A Comprehensive Survey of Unmanned Aerial Vehicles Detection and Classification Using Machine Learning Approach
- (Azeem et al., 2024): Unified multimodal fusion transformer for few shot object detection for remote sensing images
- (Li et al., 2024): From Behavior to Natural Language: Generative Approach for Unmanned Aerial Vehicle Intent Recognition
- (Seidaliyeva et al., 2023): Advances and Challenges in Drone Detection and Classification Techniques: A State-of-the-Art Review
- (Bai et al., 2023): Toward Autonomous Multi-UAV Wireless Network: A Survey of Reinforcement Learning-Based Approaches
- (Tan et al., 2025): A real-time and efficient detector for small object in UAV aerial images
- (Svanström et al., 2020): Real-Time Drone Detection and Tracking With Visible, Thermal and Acoustic Sensors
- (Liu et al., 2025): ESO-DETR: An Improved Real-Time Detection Transformer Model for Enhanced Small Object Detection in UAV Imagery
- (Li et al., 2024): Air target intent recognition method combining graphing time series and diffusion models
- (Svanström et al., 2022): Drone Detection and Tracking in Real-Time by Fusion of Different Sensing Modalities

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

References

- Al-IQubaydhi, N., Alenezi, A., Alanazi, T., Senyor, A., Alanezi, N., Alotaibi, B., Alotaibi, M., Razaque, A., & Hariri, S. (2024). Deep learning for unmanned aerial vehicles detection: A review. *Comput. Sci. Rev.*, 51, 100614. <https://doi.org/10.1016/j.cosrev.2023.100614>
- Azeem, A., Li, Z., Siddique, A., Zhang, Y., & Zhou, S. (2024). Unified multimodal fusion transformer for few shot object detection for remote sensing images. *Inf. Fusion*, 111, 102508. <https://doi.org/10.1016/j.inffus.2024.102508>
- Bai, Y., Zhao, H., Zhang, X., Chang, Z., Jäntti, R., & Yang, K. (2023). Toward Autonomous Multi-UAV Wireless Network: A Survey of Reinforcement Learning-Based Approaches. *IEEE Communications Surveys & Tutorials*, 25, 3038–3067. <https://doi.org/10.1109/comst.2023.3323344>
- Chen, R., Li, D., Gao, Z., Kuai, Y., & Wang, C. (2024). Drone-Based Visible–Thermal Object Detection with Transformers and Prompt Tuning. *Drones*. <https://doi.org/10.3390/drones8090451>
- Cheng, Q., Wang, Y., He, W., & Bai, Y. (2024). Lightweight air-to-air unmanned aerial vehicle target detection model. *Scientific Reports*, 14. <https://doi.org/10.1038/s41598-024-53181-2>

- Larrat, M., & Sales, C. (2025). A Multimodal Transformer Approach for UAV Detection and Aerial Object Recognition Using Radar, Audio, and Video Data. *ArXiv*, *abs/2511.15312*. <https://doi.org/10.48550/arxiv.2511.15312>
- Li, L., Yang, R., Lv, M., Wu, A., & Zhao, Z. (2024). From Behavior to Natural Language: Generative Approach for Unmanned Aerial Vehicle Intent Recognition. *IEEE Transactions on Artificial Intelligence*, *5*, 6196-6209. <https://doi.org/10.1109/tai.2024.3376510>
- Li, C., Wang, K., Song, Y., Wang, P., & Li, L. (2024). Air target intent recognition method combining graphing time series and diffusion models. *Chinese Journal of Aeronautics*. <https://doi.org/10.1016/j.cja.2024.08.008>
- Liu, Y., He, M., & Hui, B. (2025). ESO-DETR: An Improved Real-Time Detection Transformer Model for Enhanced Small Object Detection in UAV Imagery. *Drones*. <https://doi.org/10.3390/drones9020143>
- Luan, T., Zhou, S., Zhang, Y., & Pan, W. (2025). Fast Identification and Detection Algorithm for Maneuverable Unmanned Aircraft Based on Multimodal Data Fusion. *Mathematics*. <https://doi.org/10.3390/math13111825>
- Rahman, M., Sejan, M., Aziz, M., Tabassum, R., & Baik, J. (2024). A Comprehensive Survey of Unmanned Aerial Vehicles Detection and Classification Using Machine Learning Approach: Challenges, Solutions, and Future Directions. *Remote. Sens.*, *16*, 879. <https://doi.org/10.3390/rs16050879>
- Seidaliyeva, U., Ilipbayeva, L., Taissariyeva, K., Smailov, N., & Matson, E. (2023). Advances and Challenges in Drone Detection and Classification Techniques: A State-of-the-Art Review. *Sensors (Basel, Switzerland)*, *24*. <https://doi.org/10.3390/s24010125>
- Svanström, F., Englund, C., & Alonso-Fernandez, F. (2020). Real-Time Drone Detection and Tracking With Visible, Thermal and Acoustic Sensors. *2020 25th International Conference on Pattern Recognition (ICPR)*, 7265-7272. <https://doi.org/10.1109/icpr48806.2021.9413241>
- Svanström, F., Alonso-Fernandez, F., & Englund, C. (2022). Drone Detection and Tracking in Real-Time by Fusion of Different Sensing Modalities. *ArXiv*, *abs/2207.01927*. <https://doi.org/10.48550/arxiv.2207.01927>
- Tan, L., Liu, Z., Liu, H., Li, D., & Zhang, C. (2024). A Real-Time Unmanned Aerial Vehicle (UAV) Aerial Image Object Detection Model. *2024 International Joint Conference on Neural Networks (IJCNN)*, 1-7. <https://doi.org/10.1109/ijcnn60899.2024.10650543>
- Tan, L., Zhang, C., Bai, H., Liu, Z., & Li, Y. (2025). A real-time and efficient detector for small object in UAV aerial images. *Scientific Reports*, *15*. <https://doi.org/10.1038/s41598-025-22855-w>
- Ye, T., Qin, W., Zhao, Z., Gao, X., Deng, X., & Yu, O. (2023). Real-Time Object Detection Network in UAV-Vision Based on CNN and Transformer. *IEEE Transactions on Instrumentation and Measurement*, *72*, 1-13. <https://doi.org/10.1109/tim.2023.3241825>
- Zhao, F., Zhang, C., & Geng, B. (2024). Deep Multimodal Data Fusion. *ACM Computing Surveys*, *56*, 1 - 36. <https://doi.org/10.1145/3649447>